

Notes: Harris and Raviv (JF, 1996)

The Capital Budgeting Process: Incentives and Information

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1 Big picture

- **Question.** Why do firms use capital budgeting procedures with initial spending limits, formal requests for increases, audits, compromise allocations, and centralized approval, rather than simply applying the textbook NPV rule to every project?
- **Core answer.** Capital budgeting procedures are optimal responses to two frictions inside the firm: decentralized information and managerial empire-building incentives.
- **Information friction.** The division manager knows more than headquarters about the productivity of the division's capital, but headquarters can discover the truth only through a costly audit.
- **Incentive friction.** The division manager prefers larger capital allocations because larger divisions create private benefits, reputation, perquisites, or empire value.
- **Optimal process.** Headquarters sets an initial capital spending limit. The manager can accept it or request a larger allocation. Headquarters may ignore small requests, grant a compromise increase, or audit a large request and then either approve the requested allocation or allocate no capital.
- **Main distortion.** The optimal process can create underinvestment for high-productivity projects and overinvestment for low-productivity projects relative to first-best or NPV-maximizing investment.
- **Comparative statics.** Higher audit costs make the budgeting system more rigid. Larger investment opportunities and worse prior beliefs about productivity make the system less rigid and raise approval probabilities, compromise allocations, and salaries.
- **Economic takeaway.** Capital budgeting is an incentive mechanism. Spending limits are not simply inefficient rationing; they are part of a screening and auditing system that induces truthful communication from managers who want more capital.

2 Incremental contribution of the paper

- **From NPV rule to budgeting process.** The paper moves capital budgeting theory beyond project selection and explains the actual organizational procedure that governs project approval.
- **A theory of capital rationing inside firms.** It rationalizes capital spending limits as optimal incentive devices, not merely as financial constraints or managerial mistakes.
- **Combines asymmetric information and managerial incentives.** Earlier models often emphasized either communication or incentives. Harris and Raviv explicitly combine private information about productivity with managerial preference for capital.
- **Introduces costly auditing into internal capital allocation.** Headquarters can verify the manager's private information, but only at a cost. This creates a tradeoff between the value of better allocation and the cost of monitoring.
- **Explains observed budgeting procedures.** The model generates an initial budget, a request channel for increases, a minimum request size, stochastic audits, compromise allocations, and punishment after failed audits.
- **Provides empirical predictions.** The theory predicts how rigidity, approval probabilities, initial limits, compromise increases, and salaries should covary with audit costs, investment scale, prior productivity beliefs, and requested capital.
- **Connects corporate finance and organizational economics.** The paper treats the capital allocation process as an organizational mechanism inside the firm, not merely a finance rule.
- **Anticipates later internal-capital-market research.** It provides a microfoundation for why internal capital allocations deviate from NPV and why divisional budgets can be distorted by incentives and information.

3 Model objects and measurement

3.1 Players and timing

- Headquarters is risk-neutral and represents shareholders.
- A division manager has private information about the productivity of capital in the division.
- Headquarters can allocate capital directly or audit the manager's report at cost q .
- The manager may report one of three productivity types, $t \in \{1, 2, 3\}$.
- The model focuses on one budgeting episode, but its comparative statics map naturally into cross-sectional and business-cycle predictions.

Interpretation: Headquarters wants firm value; the manager wants capital. The capital budgeting system is designed to make reporting sufficiently truthful without auditing every request.

3.2 Technology and NPV

There are three capital productivity types. If technology type is t , an allocation of capital k produces NPV

$$V_t(k), \quad V_1'(k) < V_2'(k) < V_3'(k), \quad V_t''(k) < 0.$$

The NPV-maximizing capital level for type t is denoted k_t^* , with

$$k_1^* < k_2^* < k_3^*.$$

Interpretation: Higher-productivity technologies have higher marginal values of capital and therefore higher NPV-maximizing allocations.

3.3 Managerial preference for capital

When productivity is t , the manager's utility is

$$S + b_t k,$$

where S is salary and b_t is the marginal private benefit from controlling capital. The parameters satisfy

$$0 < b_1 < b_2 < b_3.$$

Interpretation: The manager's private benefit from capital creates the empire-building problem. He always prefers more capital, especially when productivity is higher.

3.4 Reporting, auditing, and allocations

If the manager reports type t , headquarters chooses:

- an audit probability α_t ;
- a no-audit capital allocation k_{0t} ;
- an audit-contingent capital allocation k_{1t} ;
- a salary S .

If an audit occurs and the report is truthful, headquarters allocates $k_{1t} = k_t^*$. If an audit reveals a false report, the project receives no capital.

Interpretation: Auditing is the disciplinary device, but it is costly. Compromise allocations make truthful reporting attractive while avoiding audits in every case.

3.5 First-best benchmark

Without asymmetric information and incentive problems, headquarters would choose the NPV-maximizing allocation. In the paper's first-best comparison, the allocation can also account for the fact that capital gives utility to the manager, so the first-best level can exceed the pure NPV-maximizing capital input. The main comparison remains: the optimal budgeting process distorts allocations relative to the benchmark.

Interpretation: The first-best benchmark clarifies the cost of information and incentive frictions. The distortions are not arbitrary; they are the price of eliciting truthful reports.

4 Models and theoretical specifications

4.1 Headquarters' problem

Headquarters maximizes expected NPV net of audit costs and salary:

$$\max_{S, \alpha_t, k_{0t}, k_{1t}} \sum_t \pi_t [(1 - \alpha_t)V_t(k_{0t}) + \alpha_t V_t(k_{1t}) - \alpha_t q - S]$$

subject to incentive compatibility, participation, and feasibility constraints.

Interpretation: The objective captures the tradeoff between better information through audits and the cost of auditing. The constraints force headquarters to make truthful reporting individually optimal for the manager.

4.2 Incentive compatibility

For each true type t , truthful reporting must dominate pretending to be any other type m :

$$S + (1 - \alpha_t)b_t k_{0t} + \alpha_t b_t k_{1t} \geq S + (1 - \alpha_m)b_t k_{0m}, \quad m \neq t.$$

Interpretation: A high-productivity manager may want to exaggerate or strategically choose reports that yield more capital. The mechanism must prevent profitable misreporting.

4.3 Participation

The manager must receive at least his outside option $\bar{u}$:

$$S + (1 - \alpha_t)b_t k_{0t} + \alpha_t b_t k_{1t} \geq \bar{u}.$$

Interpretation: Salary is chosen so that the manager is willing to work for the firm. The private utility from capital can substitute for salary compensation.

4.4 Proposition 1: audit probabilities

The optimal audit probabilities satisfy:

- $\alpha_1 = 0$: headquarters never audits reports of the lowest productivity.
- $0 \leq \alpha_2 \leq \alpha_3 < 1$ for $q > 0$.
- α_t is decreasing in the audit cost q .
- There exist thresholds $q_2 < q_3$ such that audit probabilities become zero above the relevant threshold.

Interpretation: Audits are most useful when the manager reports high productivity because false high-productivity claims are especially tempting and costly.

4.5 Proposition 2: capital allocations

If an audit occurs, headquarters allocates the NPV-maximizing level $k_{1t} = k_t^*$. If no audit occurs, capital allocations satisfy:

$$k_1^* < k_{01} \leq k_{02} \leq k_{03} < k_3^*.$$

The no-audit allocations converge as audit costs increase, and at high audit costs all reports receive the same allocation.

Interpretation: No-audit allocations are compromise allocations. They must be high enough to satisfy incentive constraints but low enough to limit overinvestment.

4.6 Corollary: overinvestment and underinvestment

Relative to the benchmark, the optimal capital budgeting procedure exhibits:

- overinvestment when realized productivity is low;
- underinvestment when realized productivity is high.

Interpretation: Low-productivity reports receive too much capital because the mechanism must make truthful reporting attractive. High-productivity reports receive too little capital when auditing is costly because headquarters dampens the reward for inflated claims.

4.7 Proposition 3: scale of investment opportunities

When the scale of investment opportunities s increases:

- audit thresholds rise;
- audit probabilities rise;
- the initial spending limit k_{01} falls;
- compromise and request-contingent allocations move closer to NPV-maximizing allocations.

Interpretation: Larger opportunities make misallocation more costly, so headquarters is more willing to audit and less willing to rely on coarse spending limits.

4.8 Proposition 4: prior probability of low productivity

When headquarters' prior probability of low productivity π_1 increases:

- audit probabilities rise;
- the initial spending limit k_{01} falls;
- higher-productivity no-audit allocations remain unchanged over some ranges;
- salary rises.

Interpretation: In bad times or low-productivity environments, headquarters must be especially careful about exaggerated claims. The budgeting process changes to discipline reports more strongly.

5 Identification and theoretical design logic

5.1 What the model identifies conceptually

This is a theoretical paper, so identification means isolating the mechanism rather than estimating causal effects. The model isolates the joint role of private information and empire-building preferences in generating spending limits, audits, and compromise allocations.

5.2 Why both frictions are necessary

- If managers had private information but no preference for capital, they would have no reason to misreport.
- If managers wanted more capital but headquarters had the same information, headquarters could simply impose the efficient allocation.
- The capital budgeting process matters because the manager has both private information and a reason to exaggerate productivity.

Interpretation: This is the key theoretical design: neither information nor incentives alone generate the observed budgeting process.

5.3 Observable implications

The theory predicts cross-sectional patterns:

- R&D, new product, and conglomerate divisions should face more rigid budgeting if auditing is more costly.
- Less competitive or more monopolistic divisions should face less rigid systems because investment scale is higher.
- Recessions should be associated with lower initial spending limits and higher approval probabilities, conditional on managerial outside options.
- Higher initial spending limits should covary with greater rigidity, lower approval probabilities, smaller compromise increases, and lower salaries.

Interpretation: The paper translates comparative statics into testable predictions about real capital budgeting procedures.

6 Section-by-section study guide

- **Introduction.** Motivates the gap between the NPV rule and actual budgeting practices. Presents the model's explanation: information and empire-building incentives.
- **Literature review.** Positions the paper relative to capital rationing, transfer pricing, organizational slack, and capital allocation models.
- **Model.** Defines headquarters, the division manager, productivity types, auditing, manager utility, and the optimization problem.
- **Results.** Derives audit probabilities, capital allocations, overinvestment/underinvestment, and comparative statics in audit costs, scale, and prior beliefs.
- **Empirical implications.** Reinterprets the mechanism as an initial spending limit, request process, audit/approval probability, and compromise allocation.
- **Conclusion and appendix.** Summarizes the theory and proves the mechanism's properties.

7 Figures, tables, and theoretical result blocks

7.1 Table I: Numerical Example

Read:

- Table I gives numerical solutions for different audit costs q .
- α_1 is always zero: headquarters never audits the lowest-productivity report.
- α_2 and α_3 decline as audit cost rises.
- No-audit capital allocations converge as audit cost rises.
- The manager's salary falls as audit cost rises in the numerical example.

Detailed interpretation: Table I shows the mechanism before the general propositions. The key pattern is that auditing is valuable only when it disciplines claims that would otherwise be exaggerated. As audits become costly, headquarters increasingly replaces information gathering with uniform or compromise allocations.

Economic message: The table illustrates why budgeting systems become rigid when auditing is expensive: headquarters cannot cheaply verify high capital requests, so it relies more heavily on spending limits and less on tailored allocations.

Table I
Numerical Example

This table presents numerical solutions to headquarters' problem for various values of audit cost, q . Each column shows, for a particular audit cost, the audit probabilities, α_1 , α_2 , and α_3 , the capital allocations when no audit occurs, k_{01} , k_{02} , and k_{03} , and the manager's salary, S . For these examples, the assumed production technologies are

$$V_t(k) = 20(k^{\theta_t} - k),$$

for $t = 1, 2, 3$, where $\theta_1 = 0.5$, $\theta_2 = 0.6$, and $\theta_3 = 0.7$. The other parameters are $b_1 = 0.002$, $b_2 = 0.003$, $b_3 = 1$, $\pi_1 = 0.1$, $\pi_2 = 0.3$, $\pi_3 = 0.6$, $\bar{u} = 0.0006$.

For these parameter values, the NPV maximizing investment levels are $k_1^* = 0.25$, $k_2^* = 0.27885$, and $k_3^* = 0.30455$. Finally, k_{1t} coincides with k_t^* for $t = 2$ and 3 , for all values of q (the value of k_{11} is irrelevant since $\alpha_1 = 0$). For the first-best solution, $\alpha_t = 1$ and k_{0t} is irrelevant for all t , $k_{11} = 0.25050$, and k_{1t} coincides with k_t^* for $t = 2$ and 3 .

q	0.00500	0.02500	0.04500	0.06500	0.08500	0.10500	0.12500
α_1	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
α_2	0.08576	0.01760	0.00000	0.00000	0.00000	0.00000	0.00000
α_3	0.16248	0.09823	0.06541	0.04335	0.02049	0.00000	0.00000
k_{01}	0.25437	0.27088	0.27765	0.28104	0.28453	0.28764	0.28764
k_{02}	0.27823	0.27574	0.27765	0.28104	0.28453	0.28764	0.28764
k_{03}	0.30372	0.30039	0.29708	0.29378	0.29048	0.28764	0.28764
S	0.00009	0.00006	0.00004	0.00004	0.00003	0.00002	0.00002

Figure 1: Table I: Numerical Example.

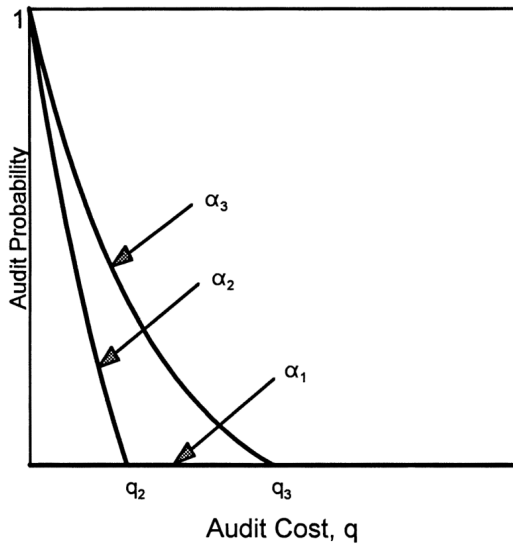
7.2 Figure 1: Audit probabilities as functions of audit cost + Figure 2: Capital allocations as functions of audit cost

Read:

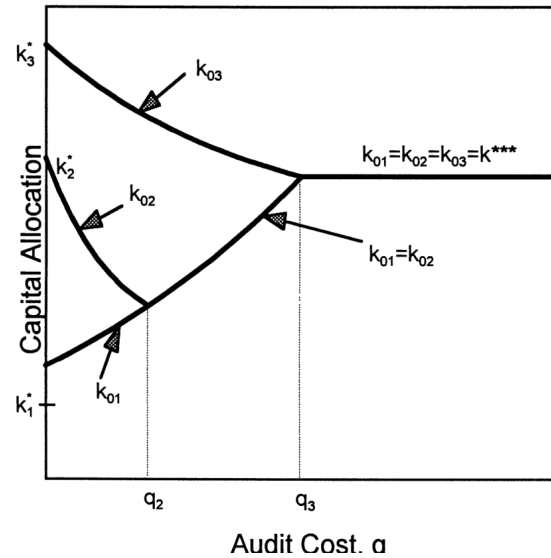
- Figure 1 shows audit probabilities falling as audit cost rises.
- The high-productivity report has the highest audit probability.
- Figure 2 shows no-audit allocations converging as audit cost rises.
- Low-productivity no-audit allocations are above the NPV-maximizing amount; high-productivity no-audit allocations are below it.

Detailed interpretation: Figure 1 and Figure 2 are the core mechanism figures. As auditing becomes costly, headquarters cannot use verification as intensely. It therefore flattens the allocation rule: different reports receive more similar no-audit allocations. That flattening preserves incentive compatibility but distorts investment.

Economic message: Audit cost is the parameter that determines whether the organization uses information-intensive budgeting or rigid capital rationing.



(a) Figure 1: audit probabilities.



(b) Figure 2: capital allocations.

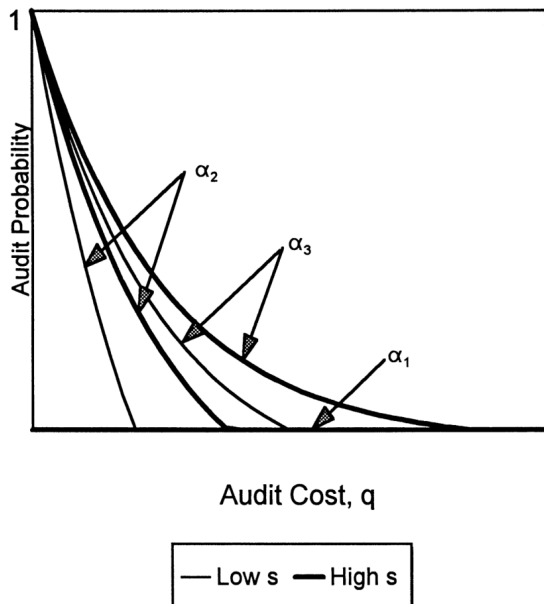
7.3 Figure 3: Effect of scale on audit probabilities + Figure 4: Effect of scale on capital allocations

Read:

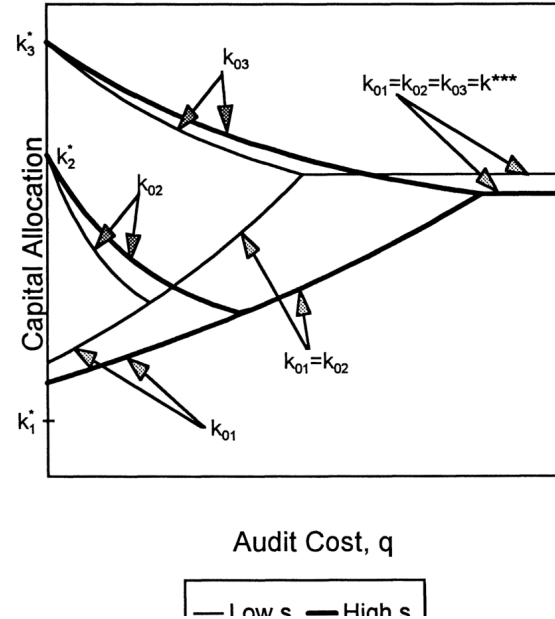
- Figure 3 compares audit probabilities when the scale of investment opportunities is low versus high.
- Audit probabilities are higher when investment scale is higher.
- Figure 4 shows that higher scale pushes no-audit allocations closer to the efficient allocations.
- Higher scale reduces the initial low-productivity spending limit and changes compromise allocations.

Detailed interpretation: Higher scale means the payoff loss from a wrong capital allocation is larger. Headquarters therefore has stronger incentives to audit and to make the budgeting process less rigid. Figures 3 and 4 show that larger investment opportunities justify more information production and more flexible allocation.

Economic message: Divisions with large or strategically important opportunities should face less rigid capital budgeting because mistakes are more costly.



(a) Figure 3: scale and audit probabilities.



(b) Figure 4: scale and capital allocations.

7.4 Figure 5: Effect of prior beliefs on capital allocations

Read:

- Figure 5 studies changes in the prior probability of low productivity, π_1 .
- Higher π_1 reduces the low-productivity no-audit allocation.
- Higher π_1 can raise audit probabilities to keep managers truthful.
- Higher-productivity no-audit allocations may remain unchanged over some ranges.

Detailed interpretation: A higher prior probability of low productivity corresponds to a worse investment environment. Headquarters becomes more worried that capital requests are exaggerated. The optimal response is to reduce the initial spending limit and audit more in the relevant ranges.

Economic message: In bad times, headquarters should tighten initial budgets but also become more willing to scrutinize and approve justified increases.

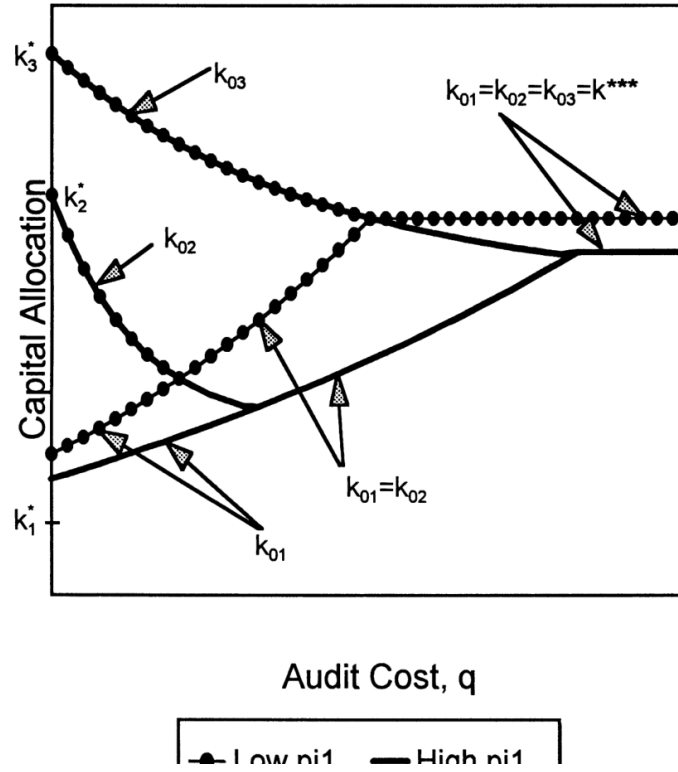


Figure 4: Figure 5: prior beliefs and capital allocations.

7.5 Table II: Direction of Change in Endogenous Variables + Table III: Covariances Between Endogenous Variables

Read:

- Table II summarizes comparative statics for rigidity, initial spending limits, approval probabilities, compromise increases, and salary.
- Higher audit cost raises rigidity and initial spending limits while lowering approval probabilities, compromise increases, and salaries.
- Higher scale lowers rigidity and initial limits while raising approval probabilities, compromise increases, and salary.
- Table III summarizes predicted covariances among endogenous budgeting outcomes.

Detailed interpretation: These tables convert the model into empirical implications. Table II is useful for thinking about exogenous differences across divisions or time periods. Table III is useful when exogenous variables are unobserved but endogenous budgeting outcomes can be jointly observed.

Economic message: The model predicts an entire budgeting system, not just one outcome. Rigidity, spending limits, approvals, compromise increases, and salary should move together in systematic ways.

Table II
Direction of Change in Various Endogenous Variables Resulting from an Increase in the Given Exogenous Variables

Endogenous Variables	Exogenous Variables			
	Audit cost, q	Scale, s	Pr(low productivity), π_1	Request, k^*
Rigidity (minimum considered request)	+	−	−	
Initial spending limit	+	−	−	
Approval probabilities	−	+	+	+
Compromise increases	−	+	0	+
Salary	−	+	+	

(a) Table II: comparative statics.

Covariances Between Pairs of Endogenous Variables

	Rigidity (Minimum Considered Request)	Initial Spending Limit	Approval Probabilities	Compromise Increases
Initial spending limit	+			
Approval probabilities	−	−		
Compromise increases	−	−	+	
Salary	−	−	+	+

(b) Table III: covariance predictions.

7.6 Theoretical Result Block 1: Capital rationing as an incentive mechanism

- Initial spending limits prevent managers from automatically receiving the capital they prefer.
- Requests for increases allow managers with high-productivity projects to communicate private information.
- Audits discipline exaggerated claims.
- Compromise allocations economize on audit costs while preserving truthful reporting.

Interpretation: The model explains why actual budgeting procedures are more complex than the NPV rule. The procedure is a mechanism for eliciting and verifying information.

7.7 Theoretical Result Block 2: Distortion pattern

- Low-productivity projects can receive too much capital.
- High-productivity projects can receive too little capital.
- This distortion is not accidental; it is required to make truthful reporting incentive compatible.
- The distortion is smaller when audit costs are lower.

Interpretation: The model delivers a nuanced view of internal capital markets: overinvestment and underinvestment can coexist inside the same optimal budgeting process.

8 Detailed result map

- **Result 1:** Optimal budgeting uses an initial spending limit.
- **Result 2:** Managers can request higher capital allocations.
- **Result 3:** Headquarters audits high reports with positive probability when audit costs are not too high.
- **Result 4:** Audited truthful reports receive NPV-maximizing capital.
- **Result 5:** No-audit allocations are compromises that support incentive compatibility.
- **Result 6:** Low-productivity projects can be overinvested and high-productivity projects underinvested.
- **Result 7:** High audit costs make budgeting more rigid.
- **Result 8:** Higher scale makes budgeting less rigid and raises approval probabilities.
- **Result 9:** Worse prior productivity beliefs lower initial spending limits and increase scrutiny.

9 Short summary

- The paper studies internal capital budgeting with decentralized information and managerial empire-building incentives.
- Headquarters wants value maximization, but the manager knows productivity and likes more capital.
- Headquarters can audit reports but auditing is costly.
- The optimal process sets an initial spending limit, permits requests for increases, and sometimes audits those requests.
- If a request is justified by audit, headquarters grants the requested capital. If the request is unjustified, no capital is allocated.
- If no audit occurs, headquarters may grant a compromise allocation between the initial limit and the requested amount.
- The optimal process can generate both overinvestment and underinvestment.
- Higher audit costs imply more rigid budgets.
- Larger investment opportunities imply more flexible budgets and higher audit probabilities.
- The paper provides testable predictions about budgeting rigidity, approval rates, compromise allocations, and salaries.

10 Conclusion

10.1 Core conclusion

Harris and Raviv show that capital budgeting procedures can be understood as optimal mechanisms for dealing with private information and managerial incentives. Spending limits, audits, and compromise allocations are not ad hoc deviations from the NPV rule. They are part of an incentive-compatible internal capital allocation system.

10.2 Economic intuition

A division manager knows more than headquarters but wants more capital. Headquarters cannot simply trust reports and cannot always audit. The optimal solution is to ration capital initially, allow larger requests, and use auditing probabilistically to discipline exaggeration.

10.3 Incremental contribution

The paper's incremental contribution is to provide a formal theory of the capital budgeting process itself. It explains observed budgeting features that standard NPV theory treats as irrelevant: initial spending limits, formal requests, approval probabilities, audit thresholds, and compromise allocations.

10.4 Implications for empirical work

The model suggests testing capital budgeting systems by relating rigidity, approval rates, spending limits, and salaries to audit costs, investment scale, and prior beliefs about productivity. It also predicts systematic covariances among endogenous budgeting outcomes even when exogenous variables are unobserved.

10.5 Limitations

The model is static and has one division with three productivity types. It abstracts from managerial compensation beyond straight salary, multi-project budgeting, repeated interactions, rolling budgets, and project-choice manipulation. These limits make the model tractable but leave important extensions.

10.6 Takeaway

The capital budgeting process is an organizational control system. It trades off the benefits of using managers' private information against the cost of verifying that information and the risk of empire-building.